

ZTA Activity Guide

A 60-minute team design clinic · 4 roles · 5 use cases (3 required + 2 bonus). Print this guide and hand one to each team.

The customer & the ask

The customer: a distributed enterprise — corporate HQ and on-prem data centre, remote branches on SD-WAN, a growing hybrid workforce (some running AI agents), and applications across on-prem, AWS and Azure. Grown by acquisition, mid-flight into public cloud; security controls haven't kept pace.

The ask: bring Zero Trust to the environment that already exists — **segment it, see it, and keep enforcing policy** wherever the user, device or workload is.

The environment

Network at a glance

- **Campus HQ** — access/distribution/core switching, wired + wireless, large IoT footprint.
- **Branches** — wired + wireless over SD-WAN.
- **Data center** — on-prem apps plus workloads in AWS/Azure.
- **Wireless** — corporate SSIDs and a Guest SSID.
- **Identity & policy** — Cisco ISE for auth, profiling and posture.

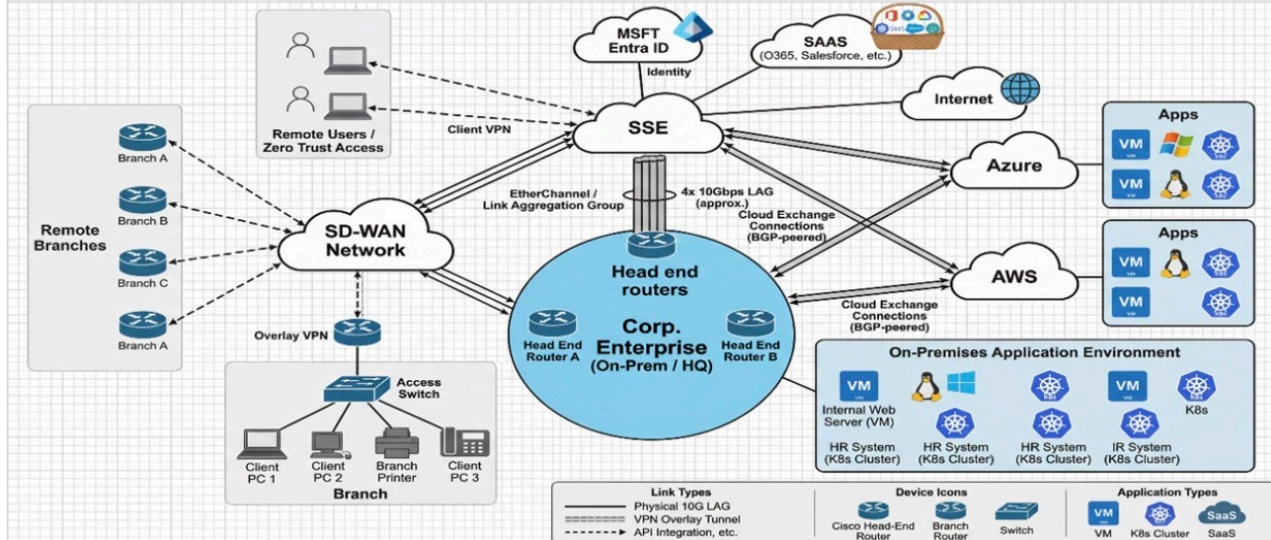
Who's on the network

- **Employees & trusted devices** (Group=Employees).
- **IoT** — printers, cameras, badge readers (Group=IOT).
- **Deskside agents** — Mac-minis (Group=deskagents).
- **Guests** — Internet-only (Group=Guest).
- **Remote workers & contractors**; some running AI agents.

Reference architecture

Everyone starts from the same picture. Your diagram for each use case is your version of this with the products and policy you propose drawn on top.

Architecture Design Clinic - Reference Network Diagram



Four roles, no passengers

OWNS THE PACE

IT Director

Watches the clock and keeps the team moving; calls time checks; confirms each use case is submitted before moving on.

OWNS THE CHALLENGE

Digital Resiliency Officer

Challenges each decision (“what would make this fail?”) to spark dialogue, surfaces the threat surface, makes the team justify trade-offs — not to block.

OWNS THE DESIGN

Network Architect

Holds the pen and draws the topology; ensures every product/control has a home on the diagram; decides placement and trust boundaries.

OWNS THE WRITE-UP

Network Security Engineer

Captures decisions; owns the pain → product mapping and product write-up; runs the diagram upload and final submission.

How the hour runs

- **Teams of 4** — one person per role; your 60-minute clock starts once all four seats are filled.
- **Five use cases, in order** — **1–3 required, 4–5 bonus**; solve Use Case 1 first, submitting one unlocks the next.
- **Required first, then bonus** — complete the three required cases, then bank the bonus rounds for extra points.
- **For each use case** — map pain → product, list products (how & why), draw a topology diagram, then submit as a team.

Segmentation methodology

Shared vocabulary — reference, not the answer.

Why VLANs / subnets alone struggle

Identity-based (group) segmentation

- Segmentation tied to IP/VLAN means topology changes ripple everywhere.
- Access rules become long IP ACLs that are hard to read, audit and maintain.
- Every new device type needs new subnets and rules; staff can't keep up.

- Classify each user/device into a group at the point of access (authentication + profiling).
- Carry the group as a tag with the traffic, independent of IP / VLAN / topology.
- Write policy as group-to-group, enforced across the existing network.

The building blocks

- Authenticate & profile every endpoint (802.1X, MAB, web auth).
- Posture — check endpoint health (AV, OS, patches) and remediate before full access.
- Tag & enforce — assign a group tag; switches / routers / firewalls enforce inline.
- Share context — feed identity/group to firewalls, DC, SD-WAN and threat tools.

How to approach each use case

- 1 · Map pain → product. For every pain point, name the product/feature that fixes it.
- 2 · Say how & why. One line on how it works and why it fits (differentiators welcome).
- 3 · Read the task & parameters. Note the specific groups, access rules and constraints.
- 4 · Propose a diagram. Draw the topology with your solution on it, then submit.

The use cases

Use Case 1 — Managed Environment — Segmentation

Required

The customer's policy is to segment campus & branch wired and wireless client devices (laptops, mobiles) into logical groups and control what each group can reach.

Customer requirements

- Segment campus & branch wired and wireless clients into logical groups: Employees, IOT, deskagents, Guest.
- Employees = laptops, mobiles (Group=Employees). IoT = printers, cameras, badge readers (Group=IoT). Deskside = dedicated Mac-minis (Group=deskagents). Guest SSID = Internet only (Group=Guest).
- IoT restricted from the other groups but needs data center and Internet access.
- Guests get Internet access only.
- Employees and IoT require controlled Internet and data center application access.
- Deskside Mac-minis reach the on-prem data center only.

Customer pain points

- Prior segmentation failed — using VLANs/IP subnets is complex and tedious.
- Profiling is labour-intensive; staff unable to keep up with constant new devices joining.
- Failing PCI audits.
- Corporate PCs not running the latest AntiVirus and OS updates.
- No budget for buying more firewalls to do segmentation.

Your tasks

- Draw a diagram of the customer's campus and branch wired/wireless networks.
- Create your proposed segmentation solution; be ready to explain the product selected, how it works, and why (include Cisco differentiators if available).
- Make sure your proposed solution addresses each customer requirement and pain point.
- Attach the diagram and submit for SME review and scoring.

Use Case 2 — Remote Workers — Zero Trust Access

Required

The customer is looking for an SSE solution to support remote workers and secure branch connectivity — moving from broad VPN to per-application Zero Trust Access, and modernizing branches with SASE.

Customer requirements

- All corporate users can use full client VPN to access selected legacy private applications — but most prefer ZTNA for modern applications.
- All corporate users need controlled internet access (block gambling and gaming).
- Contractors can only access private applications via clientless RDP access.
- Improve user experience with minimal application sign-ons.
- Customer wants to modernize branches with SASE.

Customer pain points

- Full client VPN has over-permissive privileges — not considered Zero Trust.
- User satisfaction is low due to manual VPN connection daily.
- Too many sign-ons per application created user frustration.
- Want to eliminate password logins to minimize stolen credentials.
- Existing Okta IdP is getting too expensive.
- CIO wants SASE to reduce branch costs.

Your tasks

- Draw a diagram of your proposed architecture for this use case (products, controls and policy on the reference network).
- Create your proposed solution; be ready to explain each product selected, how it works, and why (include Cisco differentiators).
- Make sure your solution addresses each customer requirement and pain point.
- Attach the diagram and submit for SME review and scoring.

Use Case 3 — AI Access and Agent Controls

Required

The customer is concerned about uncontrolled AI usage by remote employees (from Use Case 2). They want to report what's being used and build enforceable guardrails on what AI tools can do — controlling which AI tools are safe and validating information shared with approved AI sites.

Customer requirements

- Users should not be granted rights to use tools equally — fine-grained control is needed.
- Corporate users should only be able to log in to their organization's ChatGPT tenant.
- Some remote users will be running Claude Desktop agents — include controls for those employee systems.
- Users should not be able to expose sensitive data to Generative AI.
- Some users will be integrating MCP servers to local agents like Claude Code — control those integrations.

Customer pain points

- Shadow AI usage is an issue today.
- Too many unknown MCP servers for applications in both the private and public data centers.
- Users using personal credentials for ChatGPT login.
- Downloading of risky AI models.

Your tasks

- Draw a diagram of your proposed architecture for this use case (products, controls and policy on the reference network).
- Create your proposed solution; be ready to explain each product selected, how it works, and why (include Cisco differentiators).
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Use Case 4 — Merger & Acquisition — Extend Segmentation

Bonus

The customer recently acquired a small company (~100 staff) and needs to integrate it into the network quickly. The remote site has a Cisco FTD at its Internet edge and another segmenting the local AI cluster; the acquired company uses Okta as its IdP.

Customer requirements

- CISO wants to extend segmentation to the acquired company using TrustSec.
- Only selected branch users can access a restricted resource at the recently acquired company (Site 9951).

Customer pain points

- Unsure how to extend segmentation to the acquired company.
- Parent company not ready to add 100 newly acquired staff into their corporate directory — is there a quick option using what they already have?
- During transition, minimize disruptive user authentication to corporate apps and quickly secure the acquired company's apps and user population.
- Acquired company suffered an identity breach previously and is still using passwords.

Your tasks

- Draw a diagram of your proposed architecture for this use case (products, controls and policy on the reference network).
- Create your proposed solution; be ready to explain each product selected, how it works, and why (include Cisco differentiators).
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Use Case 5 — Zero Trust across Multicloud Data Center

Bonus

The customer is challenged with application sprawl across on-prem, AWS and Azure data centers. Propose a strategy and design to identify their applications (including Kubernetes) and minimize the blast radius. A comprehensive segmentation solution including application run-time protection is highly desired.

Customer requirements

- Limited budget — leverage native security features when practical; open to virtual appliances.
- CIO is seeking an end-to-end segmentation solution.
- Minimize complexity.

Customer pain points

- Minimal visibility of applications across multicloud data centers.
- Lacking a segmentation strategy and plan.
- Wide deployment of Kubernetes with latency and visibility issues.
- Unable to patch IoT devices.
- Too many disparate security management interfaces for on-prem and multicloud enforcement points.

Your tasks

- Draw a diagram of your proposed architecture for this use case (products, controls and policy on the reference network).
- Create your proposed solution; be ready to explain each product selected, how it works, and why (include Cisco differentiators).
- Make sure your solution addresses each customer requirement and pain point.
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